Data Science: Career Path Recommendations

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Data

What data are we given?
5,000 rows of data with
these information:

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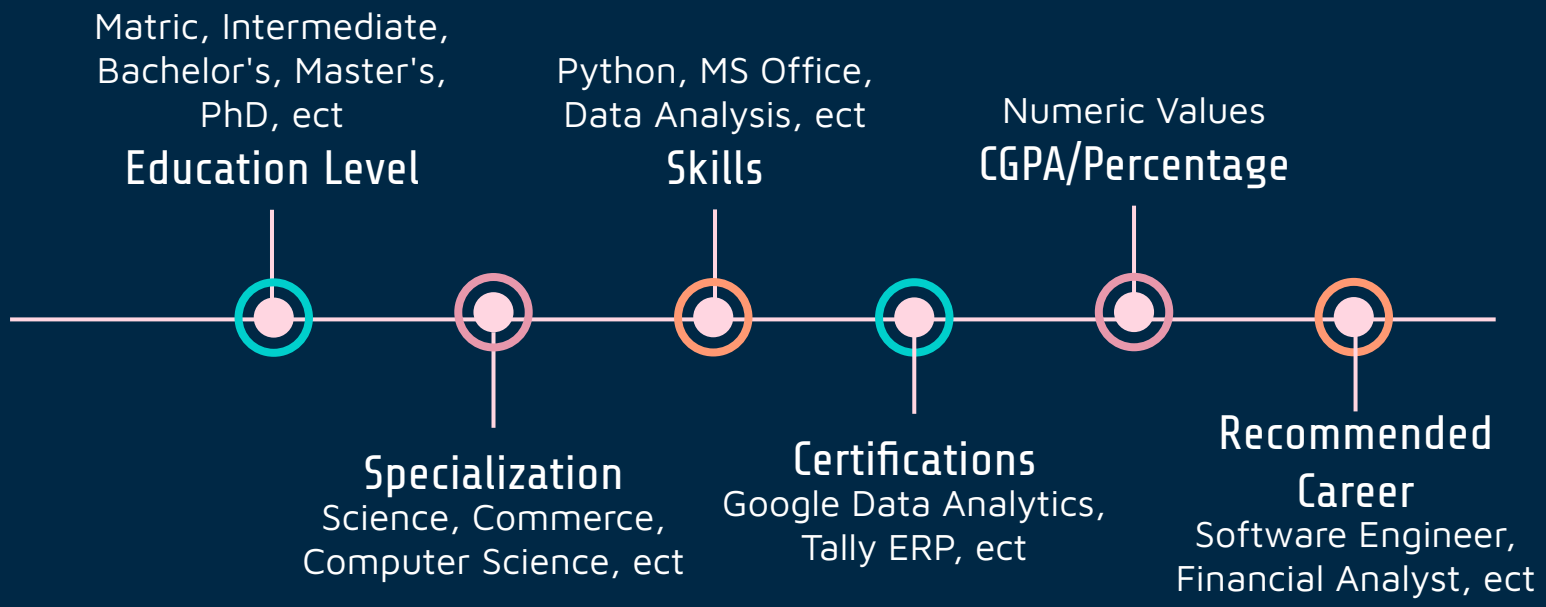
Columns


Matric, Intermediate, Bachelor's, Master's, PhD, ect
Education Level
Python, MS Office, Data Analysis, ect
Skills
Numeric Values
CGPA/Percentage
Specialization
Science, Commerce, Computer Science, ect
Certifications
Google Data Analytics, Tally ERP, ect
Recommended Career
Software Engineer, Financial Analyst, ect

Data Cleaning

What data are we missing?

Why Data Cleaning Matters

- Raw datasets are messy and unreliable
- Cleaning ensures valid statistical analysis and modeling
- EDA reveals structure, patterns, and potential issues early

Missing Values

Findings

Certifications column has 596 missing values

Missingness rate: 11.92%

All other columns: 0% missing

	Education Level	Specialization	Skills	Certifications	CGPA/Percentage	Recommended Career
count	5000	5000	5000	4404	5000.000000	5000
unique	5	8	757	7	NaN	12
top	Master's	Engineering	Accounting	Mental Health Basics	NaN	Clerk
freq	1028	668	204	674	NaN	432
mean	NaN	NaN	NaN	NaN	77.479800	NaN
std	NaN	NaN	NaN	NaN	10.396288	NaN
min	NaN	NaN	NaN	NaN	60.000000	NaN
25%	NaN	NaN	NaN	NaN	68.000000	NaN
50%	NaN	NaN	NaN	NaN	77.000000	NaN
75%	NaN	NaN	NaN	NaN	87.000000	NaN
max	NaN	NaN	NaN	NaN	95.000000	NaN

Action Taken

Filled missing Certifications using mode imputation

Ensured no column contains null values after cleaning

Handling Missing Values

Imputation Strategy

Numerical (CGPA): Filled with mean (77.47)

Categorical: Filled with mode for each column

Education Level

Specialization

Skills

Certifications

Why this works

Preserves dataset size

Avoids bias from dropping rows

Keeps categorical distributions stable

Duplicate Detection

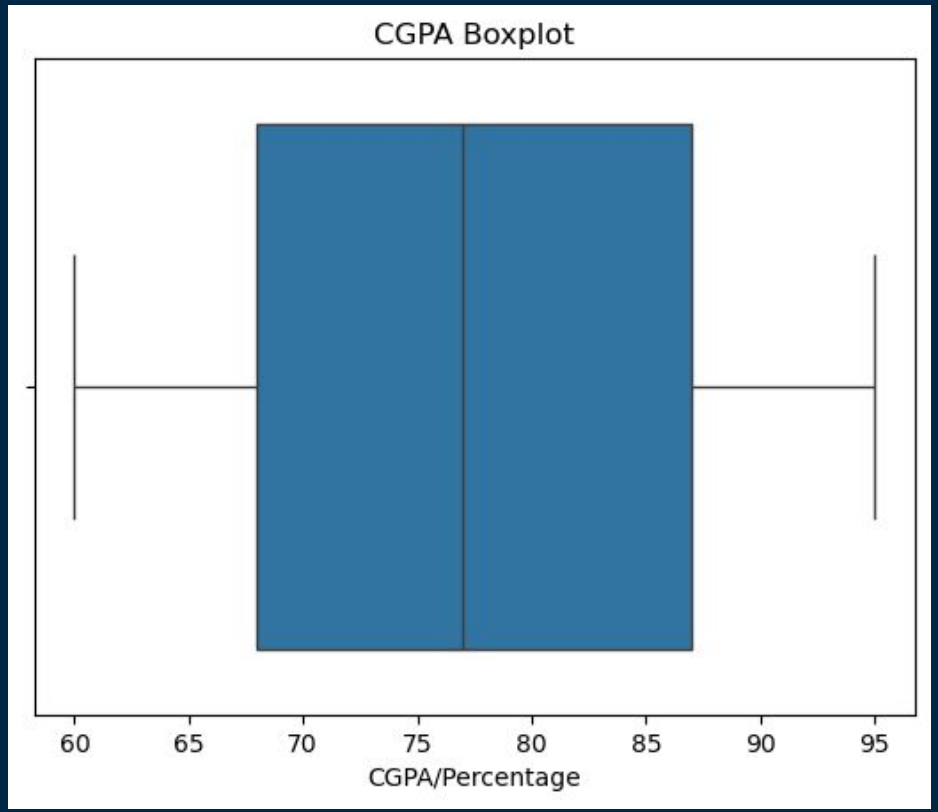
Results:

- Found 2 duplicate rows
- Removed duplicates → final dataset size: 4,998 rows

Impact:

- Ensures cleaner statistical summaries
- Prevents duplicated influence on modeling

Outlier Detection (CGPA)



Using the IQR method, we found that a small number of CGPA values fell outside bounds and were also capped as outliers using np.clip() to maintain distribution shape.

Descriptive Statistics (CGPA)

77.48

Mean

77

Median

10.39

Standard Deviation

60

Mininum

95

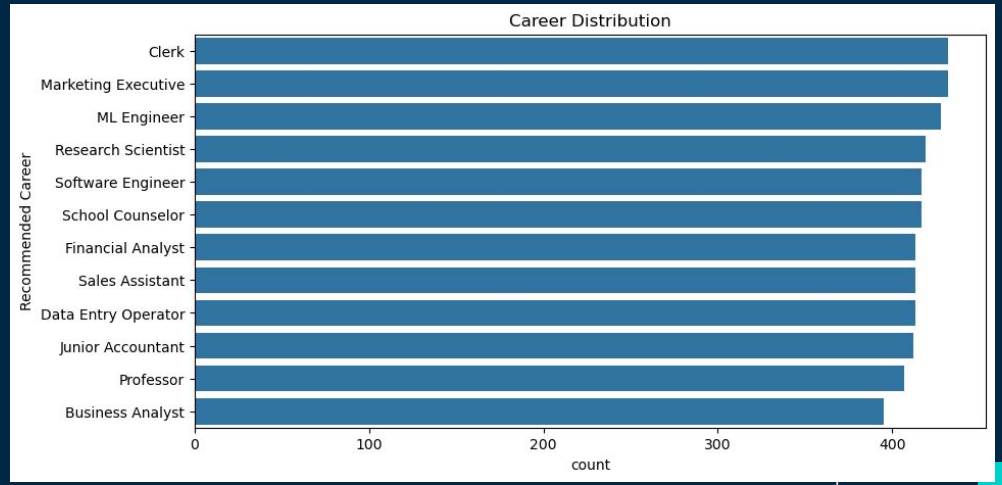
Maximum

CGPA distribution is fairly symmetric
No extreme anomalies after capping

Notes

Categorical Distributions

For recommended careers, the most common career is Clerk & Marketing while the least common is Business Analyst



Group-By Insights

Average CGPA by Career

Highest: Marketing Executive (78.56)

Lowest: Software Engineer (76.77)

Overall variation is small (~2 points)

Average CGPA by Education Level:

All levels cluster around 78

No major performance differences across education levels

Cross-Tab Analysis

All education levels appear across all careers

No single education level dominates any career

Suggests the model's recommendations are not biased toward one level

Correlation Analysis

Only one numeric column (CGPA)

No correlation matrix patterns to interpret

CGPA will still be useful for modeling

Key Takeaways

Dataset is clean, complete, and balanced

Only one column had missing values (handled with mode)

Outliers in CGPA were capped to reduce distortion

Categorical variables show diverse distributions

Group-by analysis reveals small but meaningful differences

- Dataset is ready for modeling and statistical testing

With the dataset cleaned and explored, we can now move into statistical analysis and modeling

Statistical Analysis & Hypothesis Testing

What does the data mean?

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TEST OPTIONS

T-Test

Compare mean of 2 categories

Anova

Compare means with 3 or more categories

Chi^2

Test if 2 categories are related

Correlation Test

Measure association between 2 numerical values (our data only has one aka CGPA so we would have to convert a category to numerical value to use this)

T-Test

Using a T-test to compare mean CGPA/Percentage between "Software Engineer" and "Clerk" career groups. The null hypothesis states that the mean CGPA/Percentage is the same for both groups, and the alternate hypothesis states that the means are different.

Our P-value of 0.5476311241831193 is greater than 0.05, telling us that we failed to reject the null hypothesis. There is not enough evidence to support a difference in mean CGPA/Percentage between the two categories chosen. The mean average of both categories are similar.

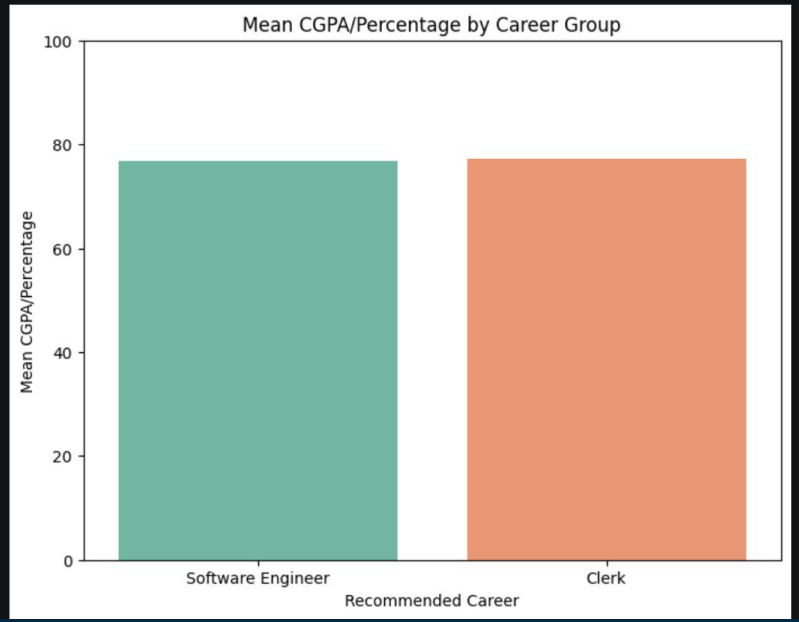
```python
# T-Test for Comparing Means of Two Groups
group1 = data[data["Recommended Career"] == "Software Engineer"]["CGPA/Percentage"]
group2 = data[data["Recommended Career"] == "Clerk"]["CGPA/Percentage"]

t_stat, p_value = ttest_ind(group1, group2, equal_var=False)

print("T-statistic:", t_stat)
print("P-value:", p_value)
```

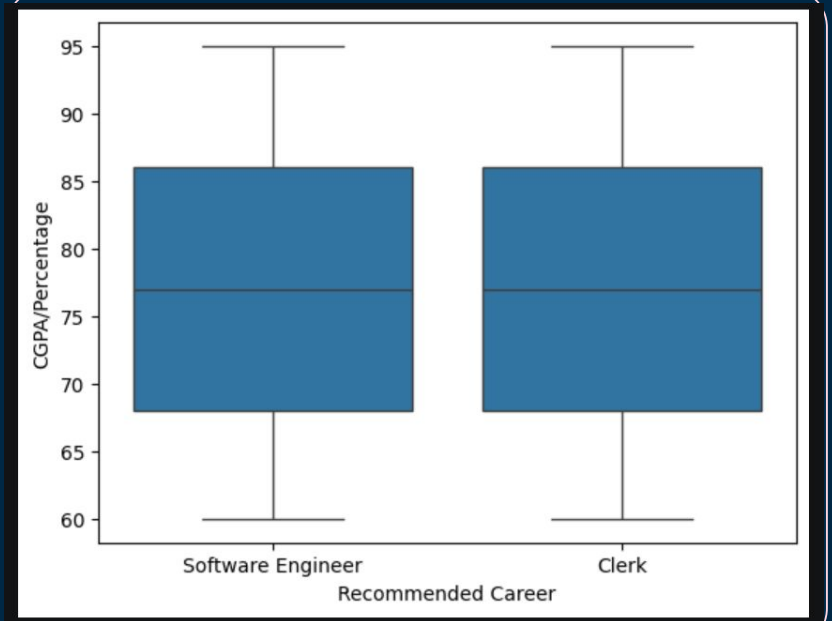
[41] ✓ 0.0s

```
T-statistic: -0.6015562346279915
P-value: 0.5476311241831193
```

T-Test Visualization

```
Software Engineer Mean: 76.77
Clerk Mean: 77.20
```





Chi-Squared Test

Using Chi-Squared test of independence to compare "Specialization" and "Recommended Career". Null hypothesis(P-value ≤ 0.05) states that the variables are independent and alternate hypothesis (P-value > 0.05) state that there is a relation.

Our P-value of 0.20192815982635112 is greater than .05 letting us know that we failed to reject the null hypothesis. There is not enough evidence to support a relation between our 2 categories. Chi-squared value by itself can't be interpreted accurately.

Statistical Analysis & Hypothesis Testing

Keep Undo

```python
# Chi-Square Test of Independence

#Imports
from pathlib import Path
import pandas as pd
import scipy.stats as stats

# Getting Dataset
data = pd.read_excel(Path("..") / "stage1" / "data.xlsx")

# Contingency table of Specilization vs Recommended Career
table = pd.crosstab(data["Specialization"], data["Recommended Career"])

# Chi-square test to test if the 2 categories are related
chi2, p, dof, expected = stats.chi2_contingency(table)

print("Chi-square Statistic:", chi2)
print("p-value:", p)
print("Degrees of Freedom:", dof)
```

✓ 0.3s Python

```
Chi-square Statistic: 84.93879861861754
p-value: 0.2507014727049615
Degrees of Freedom: 77
```

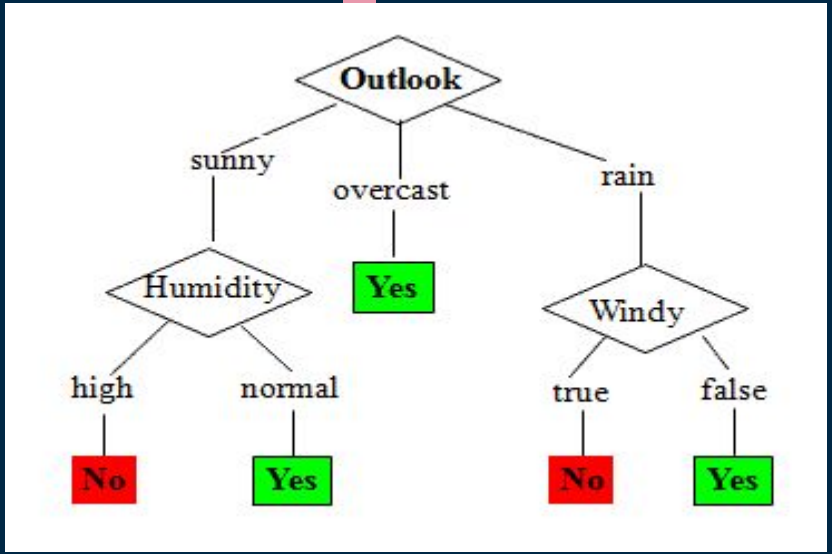
Data Modeling

What patterns can we learn from the data?

JS

Baseline Model

- For the Baseline model I decided to go with decision trees, I chose this because the data had mostly categorical features.
- Decision trees splits inputs into branches which lead to leaf nodes that give decisions(outputs)

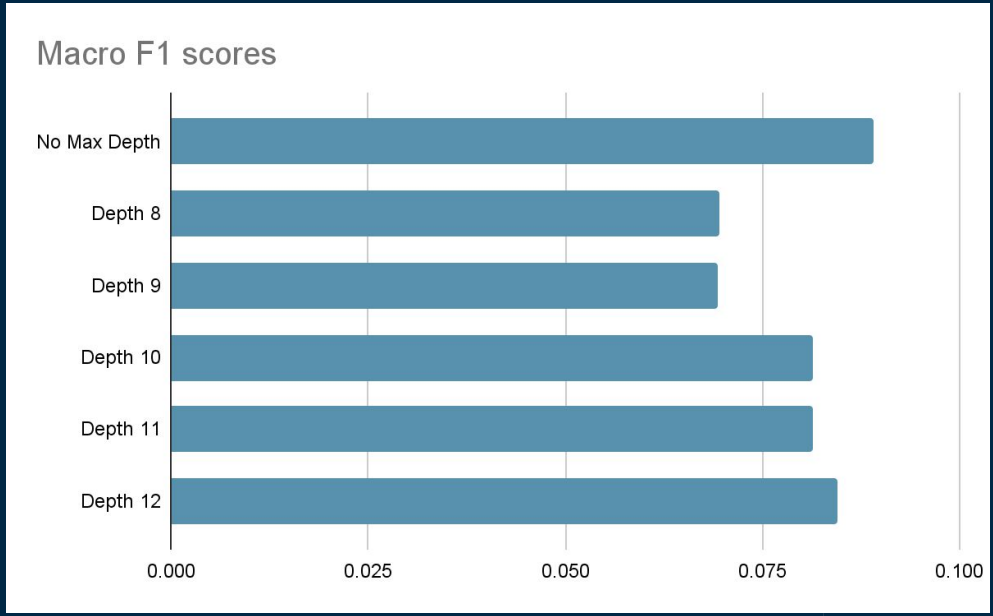


Processing The Data

- Prepared data for modeling by encoding the categorical values into numerical features
- Instead of a list of skills in a column, each skill was converted to its own column marked 1 if user had it or 0 if they didn't
- For Example skills: [Python,Ms Office, Data Analysis] -> Python:0, Ms Office: 1, Data analysis: 1

Decision Tree Result

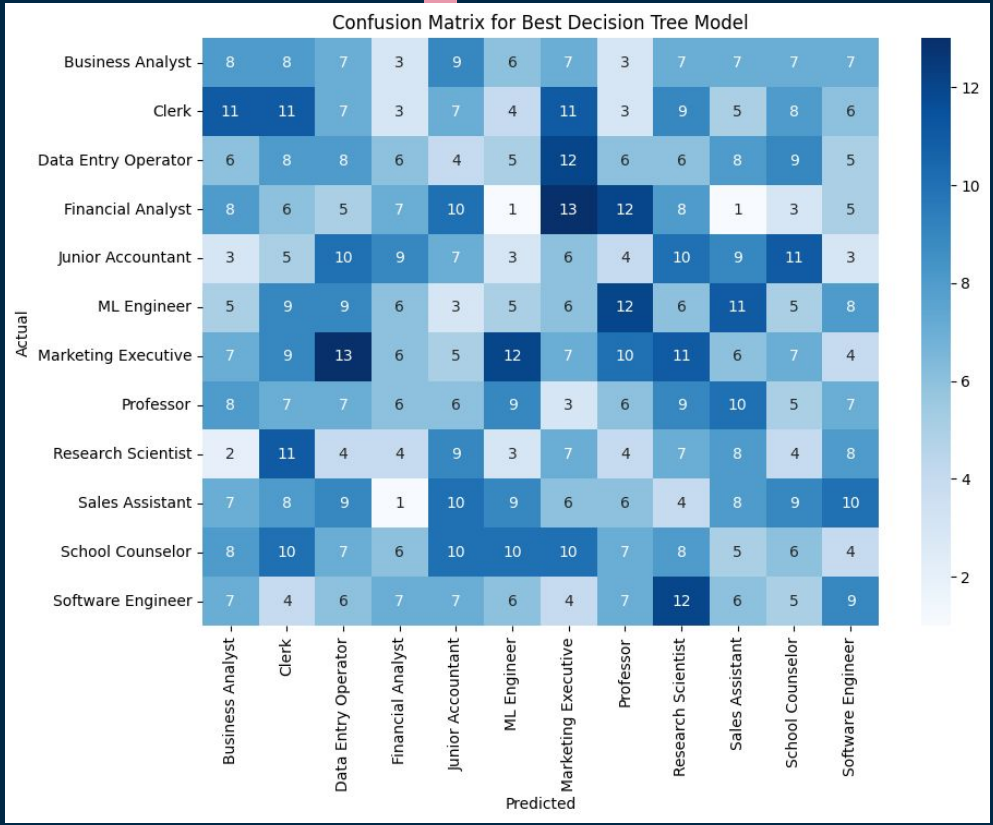
- Macro F1 scores averaged around .08
- F1 scores are how accurate and sensitive your model is with a range of 0-1 with 0 being the worst and 1 being the best
- Macro F1 scores are average F1 scores across all classes



View of the best Model

The classes on the y-axis are the actual class labels and across the x-axis are the model predictions

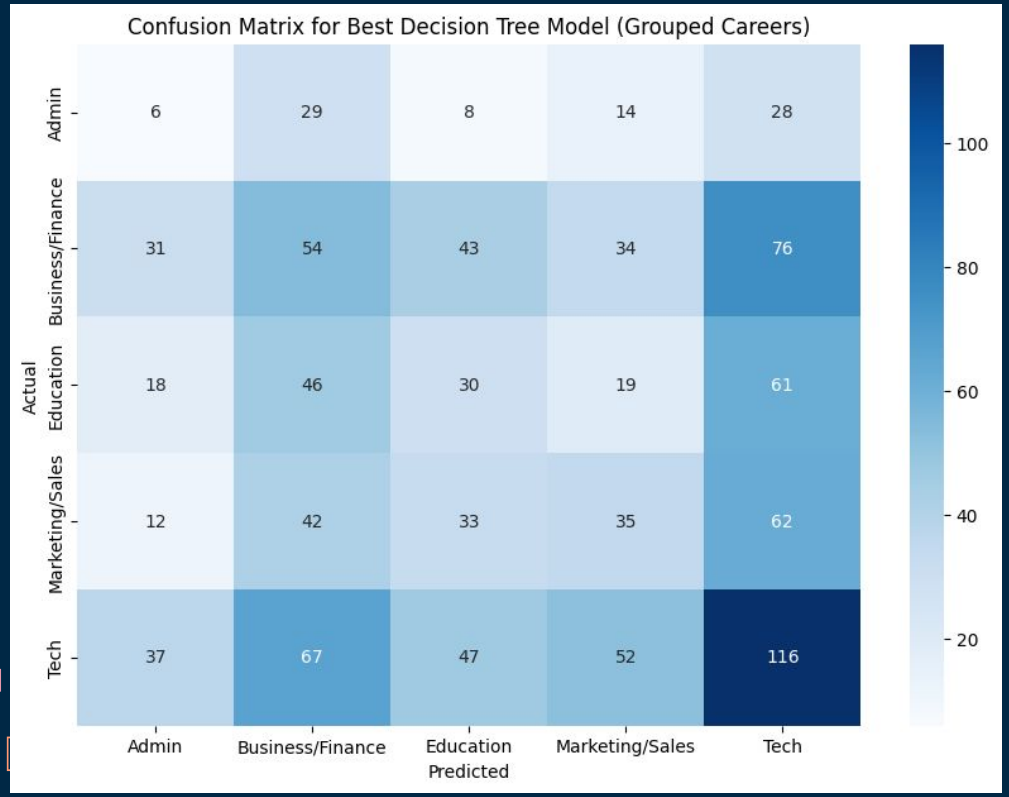
With a confusion matrix you can not only see how many times it was wrong but what the model is guessing each time for the class.



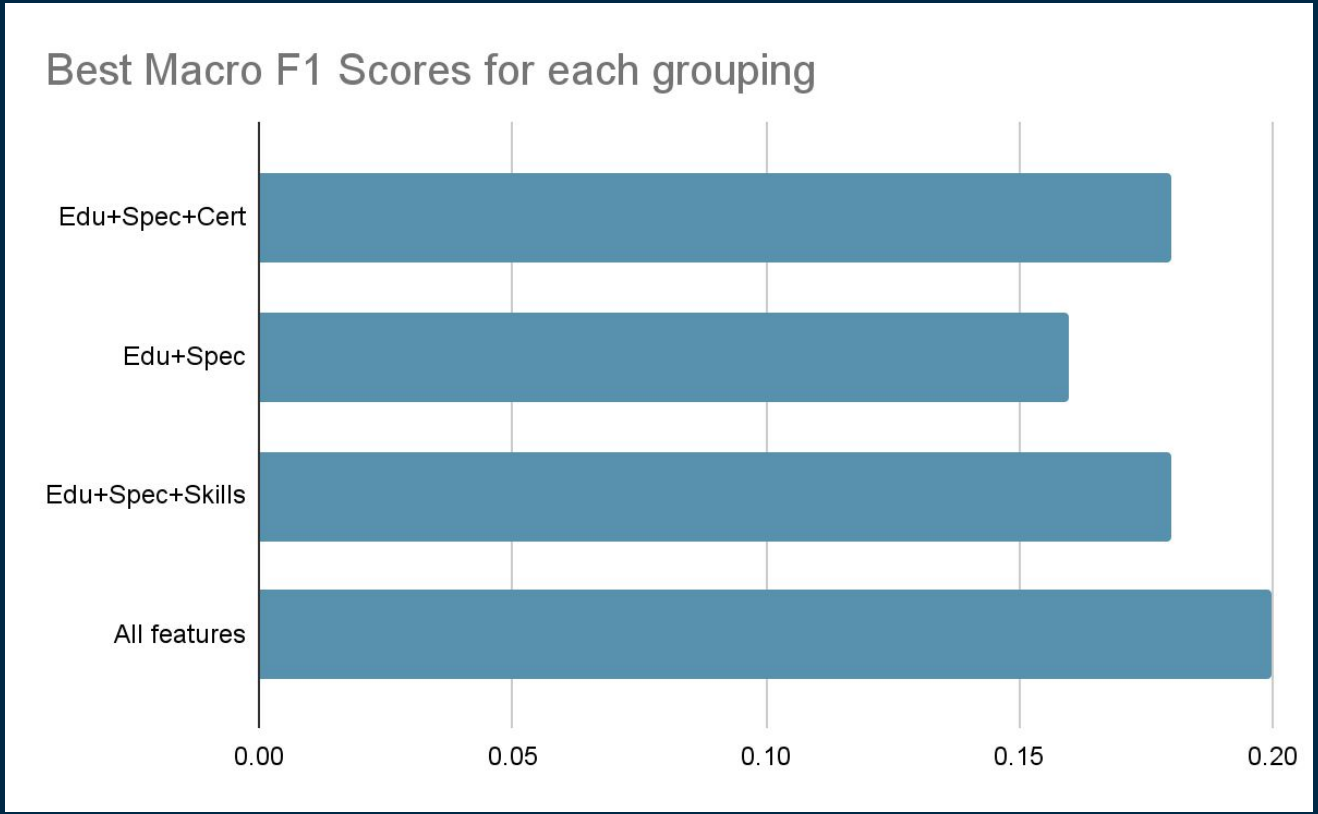
Further Data Processing

- In order to increase accuracy I processed the data more
- I reduced the different outputs from 12-5 , grouping them together
- Reduced the amount of education levels
- Used different combinations of features for decision trees

Confusion matrix for new data

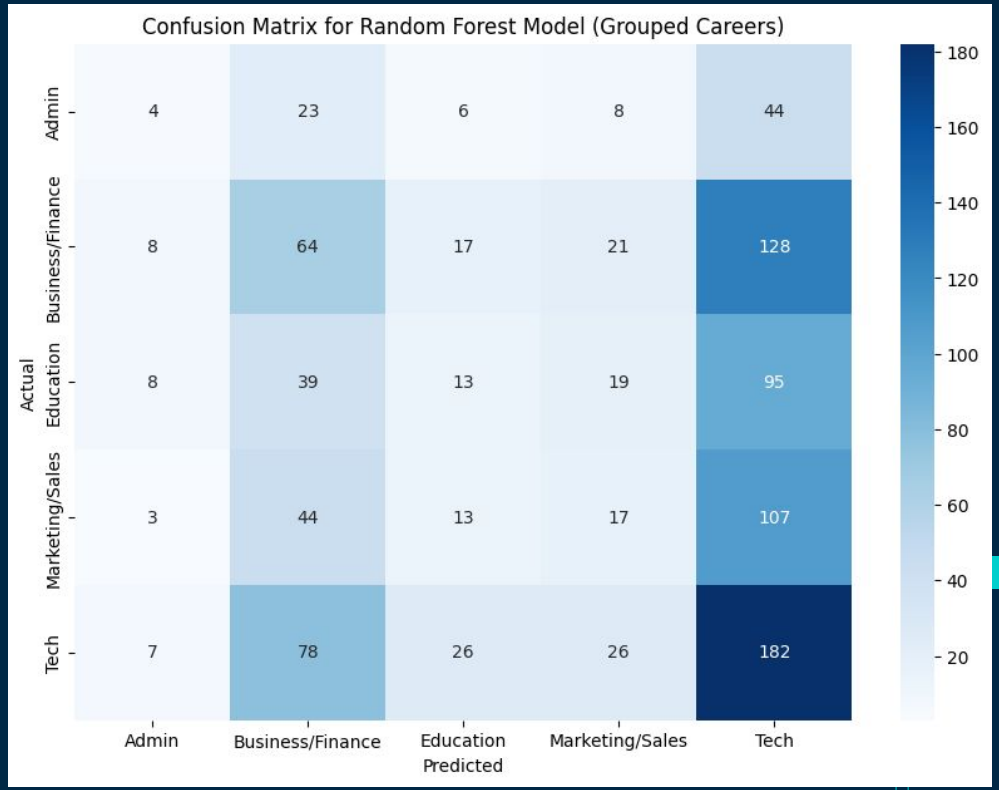


F1 Scores for different groupings



Random Forest Model

- Random forest uses multiple trees trained with bootstrap samples of the data
- Best model gave me a .19 F1 score



Information from the data

- No strong relations between different categories.
- CGPA were similar over all the different Education Level.
- Data/column did not show any strong pattern or relation to the other categories
- Couldn’t not do proper comparisons since all the data are categorical besides the CGPA

THANK YOU!

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